CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 6: Distributed Databases & NoSQL Data Modeling

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #6 | Academic Year: 2026-27

---

1. Overview & Objective:

Distributed commit protocols, CAP theorem tradeoffs, and document NoSQL design.

2. Assignment Problems & Instructions:

1. Detail the Two-Phase Commit (2PC) protocol phases and failure recovery states.
2. Explain the CAP theorem and compare consistency models of Cassandra vs MongoDB.
3. Design a document schema in MongoDB for an e-commerce platform avoiding N+1 queries.
4. Evaluate horizontal sharding strategies: range-based sharding vs hash-based sharding.

Submission Guidelines & Verification:

This document verifies end-to-end PDF storage, preview, and download for CampusNotes.
Administrators can safely edit metadata, publish/unpublish, or replace this document.